COGNITIVE·TERRAIN

BLUE vs RED · THE WAR FOR HOW A NATION THINKS



NO SHOTS FIRED. SIXTEEN WEEKS TO WIN A MIND.

The complete rules, every screen explained, and the research behind it.

8 MOVES DECODED

Plant Stories to Lessons Learned: what each card does and where it hits hardest.

SWAP CODES, NOT SERVERS

Play a friend over email or chat. No accounts needed.

EDUCATIONAL WARGAME · TURN-BASED
WINDOWS · MAC · LINUX · ANY BROWSER

CT · 0014 · 2026

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FROM THE BRIEFING ROOM

Countries compete in many ways short of war. One of them is reaching into the other side's newsfeeds, group chats and briefing rooms to change what people notice, what they believe and how fast their leaders decide.

Cognitive Terrain turns that contest into a board game you can finish over a lunch break. The board is two brains, each a network of connected nodes. Each node is a real kind of person, group or institution: a newsroom, a cabinet, a crisis team, the public's trust in its own government.

You command **Blue**, an open democracy. Your opponent is **Red**: either Olvana, a closed, centralized state, or Donovia, an oligarchic one. Over sixteen weeks you will try to cloud Red's judgment while keeping your own clear. Four crises decide who comes out ahead.

Nobody fires a shot. Nobody needs to.

FIRST GAME?

Start with **Typhoon Ines** on **Recruit** and press **START WITH TUTORIAL**. It takes about ten minutes and points at everything on this spread.

PRIMER · READ THIS FIRST

HOW A COUNTRY MAKES UP ITS MIND

Every decision runs the same loop, from a town council facing a storm to a cabinet facing a crisis. Break the loop and you break the decision.



Picture a town hearing that a storm is coming. First people **notice** the warning. Then they **interpret** it: is it serious, who says so, have we seen this before? Leaders **decide** whether to evacuate, and finally people **act**. Under all of it sits **trust**. If nobody believes the weather office, every step goes worse.

Military planners call the four steps the **OODA loop**: Observe, Orient, Decide, Act. Researchers call the fight over that loop **cognitive warfare**, a long contest over how people perceive, interpret, decide and act. In this game each brain is split into those five areas, and every node belongs to one of them.

Your score is not how many people saw your posts. It is how well each side makes decisions when it matters. That is the game's central lesson: **reach is not effect**.

THE FIVE AREAS, AND WHAT LOSING ONE COSTS YOU

When the other side holds half the nodes in one of your areas, that area is **lost** and you suffer its penalty until you win it back.

AREA	WHAT IT IS IN REAL LIFE	IF YOU LOSE IT
NOTICE	Where people get information: news, social feeds, reports from the ground, intelligence briefings.	Too much noise: you can no longer see the other side's brain clearly.
INTERPRET	How people make sense of it: who is to blame, what it means, what it looks like from the past.	Endless arguing: every move costs 1 extra point.
DECIDE	Leaders and teams who weigh options and choose: cabinets, planners, appetite for risk, red lines.	Leaders stall: 1 fewer move per turn.
ACT	Carrying decisions out: troops, supplies, public cooperation, keeping promises.	Orders go out wrong: 1 in 4 of your moves fails.
TRUST	The slow foundation: trust in institutions, unity, shared identity, knowing where to check facts.	Official statements count half, recovery halves, and enemy posts spread further.

SETUP · FIVE MINUTES

BOOT UP & DEPLOY

INSTALL

SYSTEM	WHAT TO DO
WINDOWS	Run **CognitiveTerrain-0.15-Setup.exe**. If Windows says "protected your PC", choose *More info* → *Run anyway* (the playtest build is not code-signed yet). No admin rights needed.
MAC	Unzip the Apple Silicon or Intel build, drag the app to Applications, then right-click → *Open* the first time.
LINUX	Unpack the .tar.gz and run `./cognitive-terrain`.
ANY PC	Double-click the .html file. It opens in your browser. Nothing to install.

The game works offline and never needs an account. Settings, story history and saved two-player games stay in the browser or app you play in.

THE START SCREEN

1. **Scenario.** Eight fictional campaigns: four against Olvana, four against Donovia. Typhoon Ines has no special rules, so it is the best first game. See pages 20 and 21.
2. **Language. Plain** uses everyday words ("Decision Health", "Plant Stories"). **Military** uses doctrine terms ("DQ", "Seed Channels"). Switch any time.
3. **Difficulty. Recruit**: the opponent is slower and sloppier. **Standard**: a steady, sensible opponent. **Veteran**: plans two moves ahead and plays to win. No extra budget, no hidden information: just better play.
4. **Start.** Pick the tutorial, skip straight to the game, read the in-game guide, or choose **TWO PLAYERS**.



FIG. 1 The start screen. Scenario, language, difficulty and sound are all set here before week 1.

ANATOMY OF THE WAR ROOM

EVERYTHING ON SCREEN, NUMBERED

The big brain in the middle is the one you are looking at; the small one, top right, is the other. Click either tab or the small brain to swap.



FIG. 2 Week 6 of a Typhoon Ines campaign, looking at Olvana's brain. The top-right panel lists Blue's own nodes under most pressure, starting with Social Media; below it, Looking ahead previews the next crisis.

1 TOP BAR
Week, crisis points so far (Blue – Olvana), and the next crisis with the area it will test.

2 VIEW TABS
INFLUENCE OLVANA shows their brain. PROTECT BLUE shows yours; the ▲ count is how many of your nodes are under pressure.

3 HEALTH BARS
Each side's Decision Health, 0 to 1, with marks at the crisis cut-offs. Label reads SOUND, DELAYED or ERROR.

4 BRAIN TITLE
Whose brain this is, its health, and a one-line hint about what you can do here right now.

5 THE BRAIN
Nodes are people, groups and institutions; lines are who influences whom. Faint area names mark the five regions.

6 ENTRY POINT ◇
News Media, Social Media and Government Contacts. Every attack starts here or next to a node you already lean.

7 TOP LEADERSHIP
The big core node. Worth 2 budget a turn to whoever holds the other side's.

8 THE OTHER BRAIN
A live mini-map. On your own brain it lists the nodes under pressure and suggests a defence.

9 WHAT IS HAPPENING
Looking ahead to the next crisis, health, the five areas and the event log.

10 BUDGET & MOVES
Points to spend and moves left this turn. Hover a card to see what it would cost.

11 TIMELINE
The sixteen weeks. ◆ marks a new technology, CRISIS the four decision points. Above it, this week's orders stack up as icons; click one for its story.

12 ATTACK CARDS
Four moves that push the other brain. Keys 1 to 4.

13 DEFENCE CARDS
Four moves that protect your own brain. Keys 5 to 8.

14 END TURN
Pulses when you are out of moves or budget. In two-player games it hands you a turn code.

ANATOMY · CLOSE-UPS

THE INSTRUMENTS, UP CLOSE

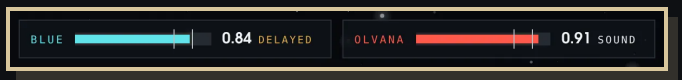


FIG. 3 Health bars. Blue on the left, Olvana on the right. The tick marks are the crisis cut-offs: above the right tick a crisis goes SOUND, between the ticks DELAYED, below them ERROR.



FIG. 4 View tabs and brain title. The "2 ▲" on PROTECT BLUE says two of your nodes are being pushed. The hint line changes with the card you pick.



FIG. 5 Budget and moves. 18 of 18 points, 3 moves left, and +6 arriving next turn. The budget is full, so all 6 would be wasted: spend it.

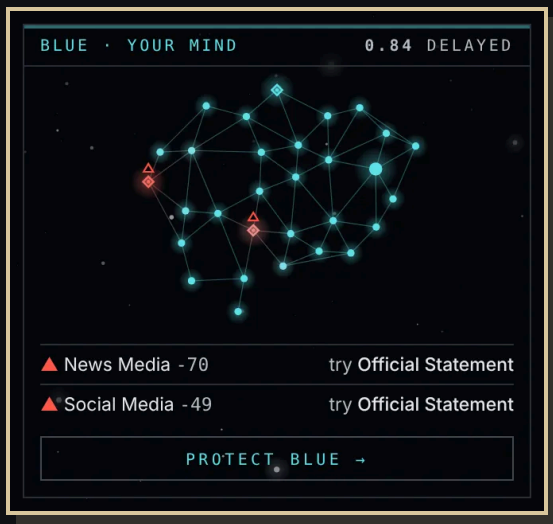


FIG. 6 The mini panel while you view Olvana. It watches your home brain for you: each ▲ row names a node under pressure and the card that would help. When one more loss would cost you an area, a red-ruled warning appears here too.

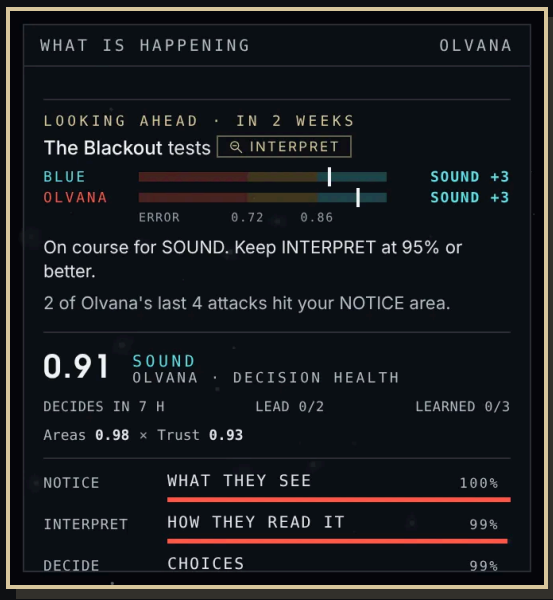


FIG. 7 What is happening. At the top, Looking ahead: the next crisis, how each side would handle it today and what would change that. Below: health, how many hours a decision takes, the race to the upper hand (LEAD 0/2) and how much of each area the brain still controls.

READING THE MAP

Each brain is a network of nodes on purpose: its nodes are people and institutions, not parts of an anatomy lesson. Colour tells you who a node leans toward; rings and small icons tell you what has been done to it.

SYMBOL KEY

Cyan node. Leans Blue. Brighter means stronger.

Red node. Leans Olvana.

Amber dotted ring. Contested: nobody holds it.

Status icons under a node. **Eye**: watched (attacks land weaker, corrections harder). **Shield**: prebunked (incoming pushes count half). **Lines**: framed (it will not recover for a while). Status words on its label use the same colours.

Dashed outline. One Watch Team or Prebunk covering a node and its links, tagged with weeks left. Pick the card and hover a node to preview what it would cover.

Large node. Top Leadership, the core of the brain.

▲ **on your brain.** Under pressure. Defend it soon.

HIDDEN BY DESIGN

Only entry points and leadership are named all the time. Hover any node to see who it is. Settings can turn names on for every node.



FIG. 8 Olvana's brain at week 6. Three entry points (◇) and Top Leadership are labelled. Olvana has prebunked its entry points and put a watch team on Social Media, so Blue has no foothold yet. The faint words are the area names.

HOLDING, LEANING AND LOSING

Every node has a control score from −100 (fully Olvana) to +100 (fully Blue). At **30** or more toward a side, that side **holds** it. At **10** or more you have a **foothold**, and your attacks can reach the nodes linked to it. Hold half the nodes in an area and the area is **lost** for its owner.

At the end of each week, nodes drift back toward their own side, and strongly held nodes push on their neighbours. A foothold you ignore will fade; a cluster you hold will spread.

WALKTHROUGH · ONE MOVE FROM START TO FINISH

PICK. AIM. PREVIEW. STRIKE.

Each week you get three moves and a budget. Here is one move, start to finish.



STEP 1 · PICK A CARD Click a card in the bottom bar or press 1 to 8. Red-family attacks on the left, defences on the right. Each shows its cost in points. Picking an attack jumps the view to Olvana; a defence jumps it to your brain.



STEP 2 · FIND A RINGED NODE With Plant Stories selected, the nodes you can reach this turn are ringed. Entry points are always in reach; other nodes need a foothold next door.



STEP 3 · HOVER TO PREVIEW The hover card shows who the node is, who holds it, and what this move would do: the typical shift, the chance to fail and the cost. The side panel adds the bonuses and the real-life reasons. Click to commit.



STEP 4 · SEE WHAT HAPPENED The headline, a few lines of the story, and what changed: here a B for spread and +27 opinion on News Media. FULL STORY opens the complete report.

PRO TIP

Each card's best areas multiply its punch by up to 1.5. Plant Stories hits hardest in **NOTICE**, Frame the Story in **INTERPRET**, Bot Swarm in **ACT**.

MOVES CAN FAIL

About 1 in 12 attacks is taken down or falls flat, more often on watched or prebunked nodes. A failed attack nudges the node *toward* the defender. Official Statements are sometimes ignored, especially once your TRUST area is lost.

BUDGET · MOVES · THE WEEKLY TICK

SPEND IT OR LOSE IT



FIG. 9 Three moves and a full budget of 18 points.



FIG. 10 Out of moves with points left over: the +1 MOVE button appears. Blue can buy one extra move a turn for 2 points.

DON'T HOARD

Your budget tops out at **18**. Anything past that is thrown away at the start of your next turn. If the bar is nearly full, surge.

WHERE POINTS COME FROM

SOURCE	PER TURN
Base income	+6
Each enemy node you hold	+1
Their Top Leadership, if you hold it	+2
Their Government Contacts entry point	+2

Most cards cost 2. Bot Swarm and Lessons Learned cost 3. Lose your INTERPRET area and everything costs 1 more. Lose DECIDE and you get only 2 moves a turn.

THE WEEK, IN ORDER

1. **New technology** (weeks 2, 6, 10, 14): both sides choose to use it or prepare for it.
2. **Blue's turn.** Collect budget, make up to 3 moves (plus a surge), press END TURN.
3. **Olvana's turn.** Each move lights up on your map, numbered, with a small card saying what it did and why.
4. **Upkeep.** Every node recovers a little toward its own side, unless framed. Strongly held nodes push on their neighbours. Lessons Learned makes your recovery faster.
5. **Crisis** (weeks 4, 8, 12, 16): both sides are scored. See page 15.
6. **Week in review.** Both sides' moves, what changed and what is next. See page 16.

WATCH THE CLOCK

The top bar always names the next crisis and the area it tests, e.g. `CRISIS 2 · 2 WK · INTERPRET`. That area is where you need to be strong in two weeks.

ATTACK CARDS · KEYS 1–4

FOUR WAYS TO GET INSIDE

Attacks go on the other side's brain. Each is a real family of influence technique drawn from the research behind the game.

PLANT STORIES

KEY 1

GET THE MESSAGE IN · MILITARY: SEED CHANNELS

COST	PUSH	TARGETS	SPECIAL
2	20	1	×1.5 ◇

NOTICE ×1.5 | INTERPRET | DECIDE ×.75 | ACT ×.75 | TRUST

Place content in the news and feeds people already use. The strongest single push in the game, and 50% stronger on an entry point. Your opening move.

> IN REAL LIFE: look-alike local news sites, sponsored posts, paid influencers.

FRAME THE STORY

KEY 2

SHAPE THE STORY · MILITARY: FRAME & PRIME

COST	PUSH	TARGETS	SPECIAL
2	16	1	FRAMED

NOTICE ×.75 | **INTERPRET ×1.5** | **DECIDE ×1.25** | ACT | **TRUST ×1.25**

Tell it in a way that triggers emotion or identity. The target stops recovering for 2 turns, so a follow-up hit sticks. The setup move before a capture.

> IN REAL LIFE: a clip edited and captioned to make people angry or afraid.

BOT SWARM

KEY 3

MAKE IT LOUD · MILITARY: COORDINATED AMPLIFICATION

COST	PUSH	TARGETS	SPLASH
3	12	+LINKS	7 EACH

NOTICE | INTERPRET | DECIDE | **ACT ×1.5** | **TRUST ×1.25**

Many coordinated accounts repeat it at once. Hits the target *and every node linked to it*. Hold their Social Media entry point and the splash hits at full strength.

> IN REAL LIFE: thousands of accounts posting the same thing within an hour.

REPOST & REBRAND

KEY 4

ADAPT · MILITARY: PIVOT & RESEED

COST	PUSH	TARGETS	SPECIAL
2	10	+LINKS	STRIP

NOTICE | INTERPRET | DECIDE | ACT | TRUST

Come back with new accounts and a new angle after removals. Strips **Watched** and **Prebunked** from the target and its links. Use it when the other side digs in.

> IN REAL LIFE: a banned network reappearing under new names the next day.

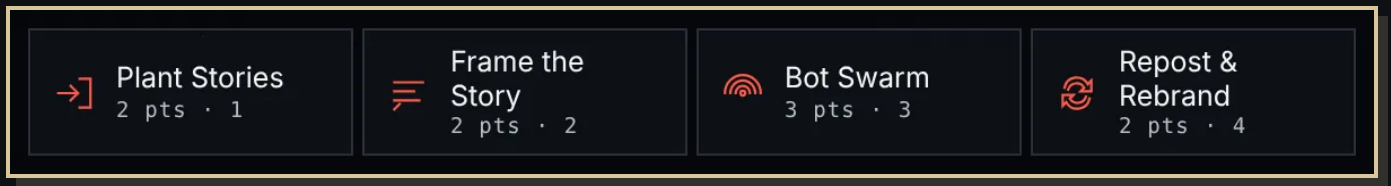


FIG. A The attack half of the card bar. Each card shows its icon, cost and hotkey; a card you cannot afford is dimmed.

COMBO

Frame the Story a node this week, then **Plant Stories** on it next week: it cannot recover in between, so the second push usually captures it. If it is prebunked, open with **Repost & Rebrand**.

DEFENCE CARDS · KEYS 5–8

FOUR WAYS TO HOLD THE LINE

Defences go on your own brain. Attacking is cheap and defending is not: the research calls it the attacker's cost advantage. Spend where the next crisis will look.

WATCH TEAM

KEY 5

SPOT IT · MILITARY: ASSESSMENT CELL

COST	LASTS	TARGETS	BOOST
2	3 WK	+LINKS	×1.5

NOTICE ×1.5 · INTERPRET · DECIDE · ACT · TRUST

Watches a node and its links. Incoming posts land a grade weaker and are more likely to be exposed, framing is cleared, and your Official Statements there hit 50% harder.

> IN REAL LIFE: analysts tracking sudden spikes of identical posts.

PREBUNK

KEY 6

BUILD IMMUNITY · MILITARY: PREBUNK & INOCULATE

COST	LASTS	TARGETS	INCOMING
2	3 WK	+LINKS	×0.5

NOTICE · **INTERPRET ×1.5** · DECIDE · ACT · TRUST

Teach people the trick before they see it. A whole cluster takes half influence for three weeks. Best laid down *before* the pressure arrives.

> IN REAL LIFE: short videos showing how fake clips are made, shown before an election.

OFFICIAL STATEMENT

KEY 7

SET THE RECORD STRAIGHT · MILITARY: AUTHORITATIVE RESPONSE

COST	RESTORE	TARGETS	IF WATCHED
2	14+	1	×1.5

NOTICE · INTERPRET · **DECIDE ×1.5** · ACT · TRUST

Correct the record through trusted voices. Pulls one node back toward you. Blue's free press makes it hit extra hard while Blue's TRUST holds, and it counts half once TRUST is lost.

> IN REAL LIFE: a press conference with evidence from a trusted spokesperson.

LESSONS LEARNED

KEY 8

LEARN & PRACTICE · MILITARY: AFTER-ACTION REVIEW

COST	ALL NODES	TARGET	MAX
3	+4	NONE	3×

NOTICE · INTERPRET · DECIDE · **ACT ×1.5** · TRUST

Review what went wrong and drill. Every node you own gains 4, framing is cleared, and your weekly recovery gets permanently faster (up to three times). Pays more the earlier you buy it.

> IN REAL LIFE: an after-action review followed by a practice exercise.

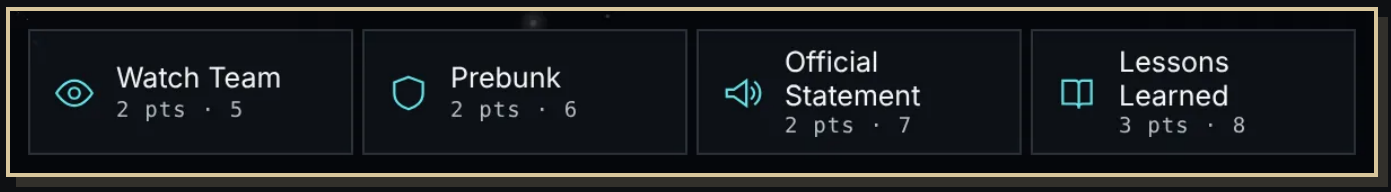


FIG. B The defence half of the card bar. Lessons Learned needs no target: click it and it fires.

PRO TIP

Watch Team first, Official Statement second, on the same node in the same turn: the statement lands 50% harder. Against a Bot Swarm wave, Prebunk the cluster instead of fixing nodes one by one.

PROTECT YOUR OWN BRAIN

Olvana never skips a turn. If you only attack, your own Decision Health will slide by week six and the crises will punish you for it.

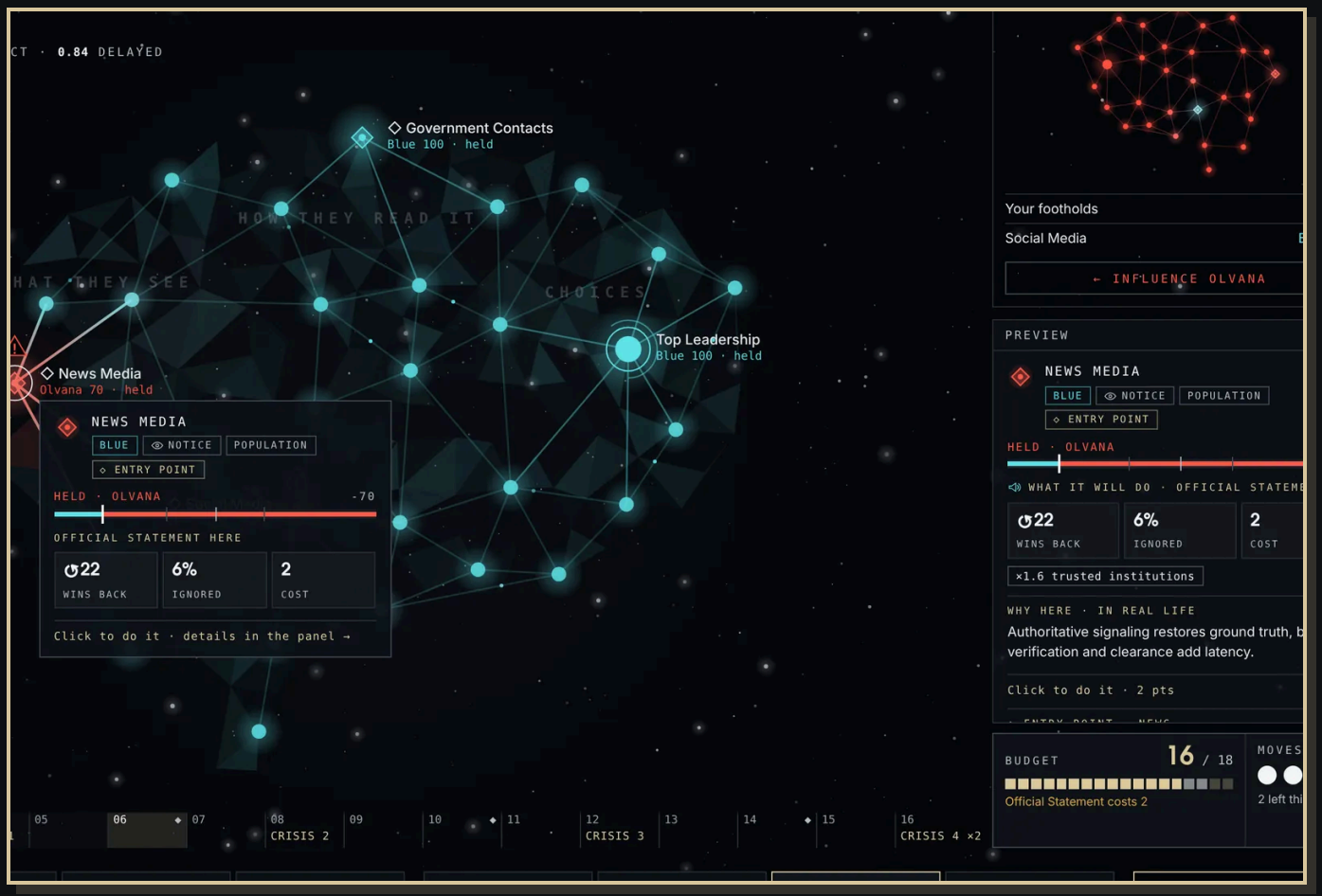


FIG. 11 PROTECT BLUE with Official Statement selected, hovering Blue's Social Media, which Olvana now holds. The hover card shows it would win back 22 points; the side panel adds the bonus from trusted institutions and why a statement works there.

Switch to **PROTECT BLUE** at least once a turn, or simply read the mini panel while you attack: it lists the three nodes in most trouble and suggests a card for each. The tab shows the count, like `PROTECT BLUE · 5 ▲`.

When an area is one node from falling, the mini panel shows a red-ruled warning naming the penalty you are about to take. That is the moment to spend, even if it means skipping an attack.

Think about **where the next crisis looks**. A crisis scores 60% on your overall Decision Health and 40% on the one area it tests. Two weeks before a crisis, put your defence there.

Of all your areas, **TRUST** is the one to guard. It falls slowly but costs the most: your statements count half, your recovery halves, and every Olvana post spreads a grade further. Blue's free press helps, because TRUST recovers faster for Blue than any other area.

THE NUMBERS BEHIND EVERY POST

VIEWS ARE CHEAP. SHARES COUNT.

Every attack is a small campaign of posts. The game rolls the audience it reached, then grades the campaign the way the research does.

SPREAD SCORE PER POST

(0.1·views + 0.3·likes + 0.7·comments + 1·shares) ÷ posts

RUSHING & XU (2025) · ENGAGEMENT EFFECTIVENESS

GRADE	SCORE	MEANING	EFFECT
A+	10,000+	Viral	×1.6
A	1,000+	Excellent	×1.4
B	100+	Above average	×1.2
C	10+	Average	×1.0
D	4–9	Below average	×0.7
E	3	Poor	×0.45
F	0–2	Failure	×0.25

Each move type has its own crowd. **Plant Stories** draws lots of views and few shares. **Frame the Story** sparks the most comments. **Bot Swarm** racks up shares but little real conversation. Better posts earn more interaction per view, so a viral post looks viral in the numbers too.



FIG. 12 The full story (◉ in the log, or FULL STORY on a result card): audience counts with their weights, the grade and its formula, what changed and why it matters.

WHAT PUSHES THE GRADE

UP	DOWN
Posting on an entry point	A Watch Team on the target (−1 grade)
Holding their News Media	Olvana's censored internet (Blue's posts)
Their TRUST area lost (+1 grade)	
A new technology you chose to use	

REACH ≠ EFFECT

A grade only multiplies the push. A viral post on the wrong area still loses the crisis. Aim first, then worry about virality.

WEEKS 2 · 6 · 10 · 14

NEW TECHNOLOGY ARRIVES



FIG. 13 The timeline. ◆ marks a new technology; CRISIS 1–4 fall in weeks 4, 8, 12 and 16, and crisis 4 counts double (×2).

Four times a campaign a new technology changes the board. Both sides choose in secret: **USE IT** for an attack bonus, or **PREPARE DEFENSES** to deny the other side's bonus and prebunk your three entry points for two weeks.

If you use it and they prepared, you get nothing. If you prepared and they used it, they get nothing against you. The choices show in the log, and in the New Technologies list in the side panel.

TECHNOLOGY	BOOSTS
ALGORITHMIC TARGETING	Frame the Story +1 grade
SOCIAL BOTS	Bot Swarm +1 grade, full splash
TROLL FACTORIES	Bot Swarm, Repost & Rebrand +1 grade
AI SYNTHETIC MEDIA	Every attack +1 grade
HUMANOID ROBOTS	Plant Stories reaches any NOTICE or INTERPRET node, no foothold needed

Each campaign uses four of these. The scoring method, the 1–3 scale and the threshold of 9 come from Rushing, Umeokolo & Xu (2025), whose forecast predicts a new disruptive innovation roughly every 4.6 years. The paper scores AI Synthetic Media and Humanoid Robots; the other score sets are the designer's estimates.

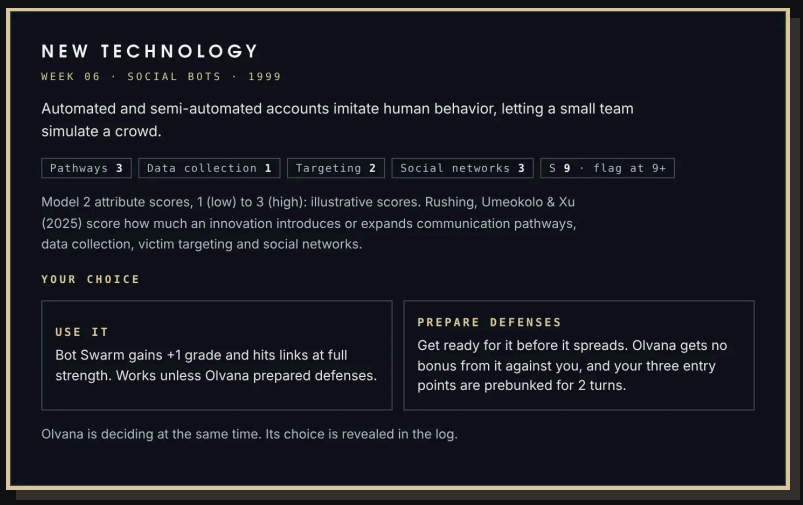


FIG. 14 Social Bots arrive in week 6. The four attribute scores (pathways, data collection, targeting, social networks) sum to S = 9 (the designer's estimate), the paper's threshold for a flagged innovation.

WEEKS 4 · 8 · 12 · 16 · HOW TO WIN

FOUR MOMENTS DECIDE IT ALL

At the end of weeks 4, 8, 12 and 16 a crisis hits the scenario's setting: a contaminated-water rumour, a severed cable, a fake evacuation alert. Each crisis tests one area, named in the top bar weeks ahead.

Each side's crisis result comes from **60% overall Decision Health + 40% of the tested area**, with an extra penalty if that area is lost.

RESULT	NEEDS	POINTS
SOUND	0.86 or better	3
DELAYED	0.72 – 0.85	1
ERROR	below 0.72	0, and the other side gets 1

Crisis 4 counts **double**. Most points after week 16 wins; a tie goes to the healthier side.

HOW DECISION HEALTH IS WORKED OUT

Each area counts by how much the side leans on it: Blue **20 / 25 / 30 / 25%** for NOTICE / INTERPRET / DECIDE / ACT; both opponents **15 / 20 / 40 / 25%**. That total is scaled by TRUST (×0.75 to ×1), then **0.06** comes off for each area lost, and any flood penalty. Difficulty never changes it. The side panel shows the sum.

THE KNOCKOUT

Keep the other side at least **two areas worse off** than you for **two weeks running** and you win on the spot. The side panel tracks it as LEAD 0/2.

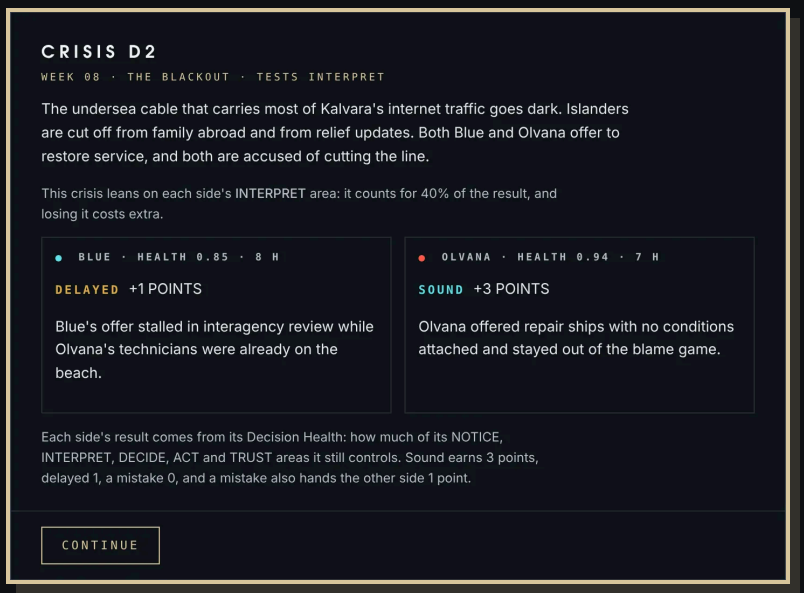


FIG. 15 Crisis 2, The Blackout. Blue was slow (DELAYED, 1 point); Olvana handled it soundly (3 points). The text says what each side actually did.

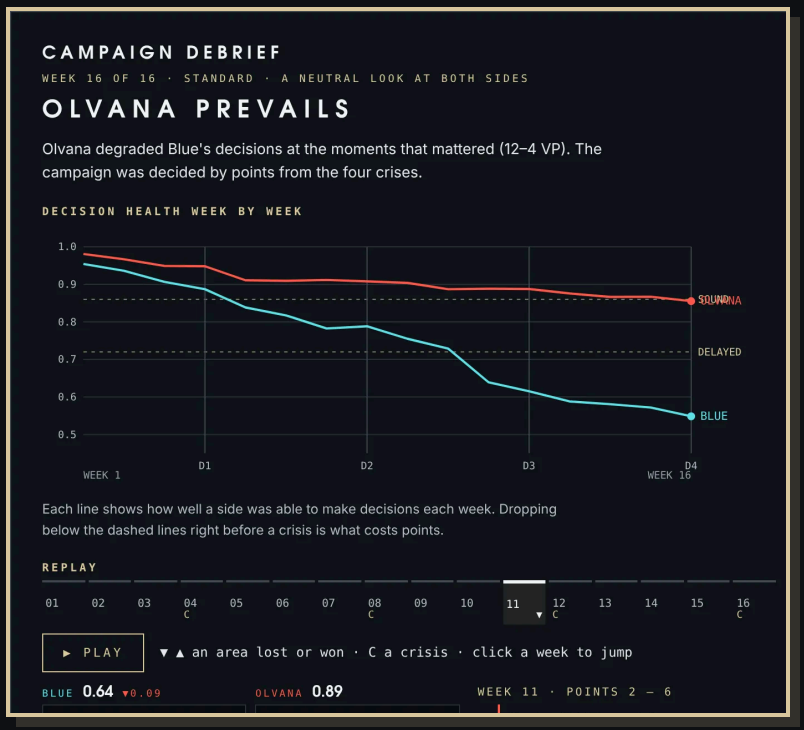


FIG. 16 The debrief: a neutral look at both sides, health week by week, three ways to judge who led, the turning points and lessons for next time.

WEEK IN REVIEW · LOOKING AHEAD · REPLAY

KNOW WHAT JUST HAPPENED

NEW IN 0.13.4

The game explains itself at three moments: while Olvana moves, at the end of every week, and after the campaign.

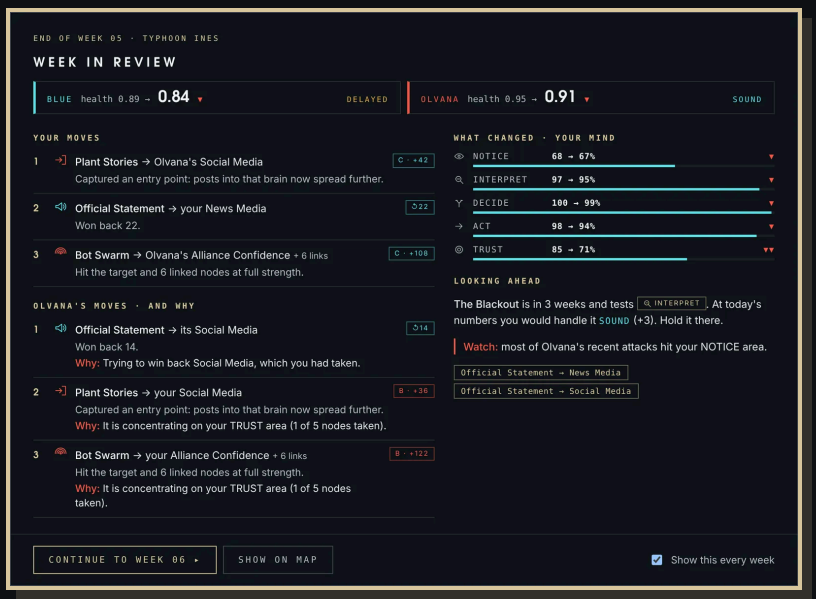


FIG. C Week in review. Both sides' health, your moves and what each achieved, Olvana's moves with the most likely reason, how each of your areas moved, and what the next crisis needs.

WHILE OLVANA MOVES



FIG. D Olvana's second move lights up the node it hit and the links it reached; the rest of the map dims. Its card, over the side panel, says why it chose that target.

AT THE END OF EVERY WEEK

Read the week in review, then **CONTINUE**, or **SHOW ON MAP** to see the nodes that changed lit up first. It can be switched off in Settings.

LOOKING AHEAD

From two weeks before a crisis, the side panel shows how each side would handle it today, how strong the tested area needs to be for SOUND, and where Olvana has been aiming. It judges Olvana only from moves you have seen.

USE IT

If Looking ahead says DELAYED, put this turn's defence into the tested area. Two weeks is just enough time.

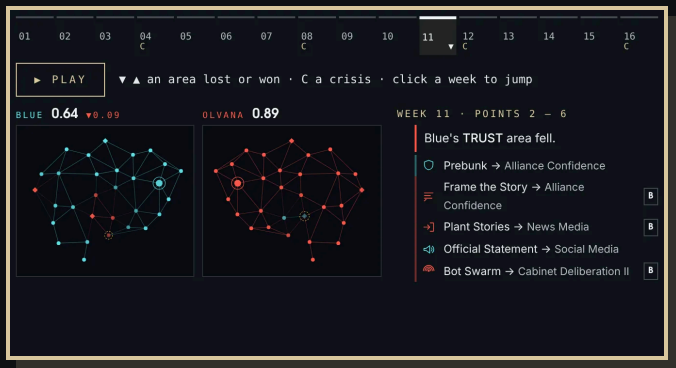


FIG. E The debrief replay. Step or play through the campaign: both minds week by week, the moves that week, and the turning points (▼ ▲) and crises (C) on the strip.

HEAD TO HEAD · NO SERVER REQUIRED

PLAY A FRIEND BY SWAPPING CODES

NEW IN 0.13

Each turn ends with a short line of text. Send it any way you like: email, chat, Teams. Your opponent pastes it in, watches your moves play out, takes their turn and sends one back.

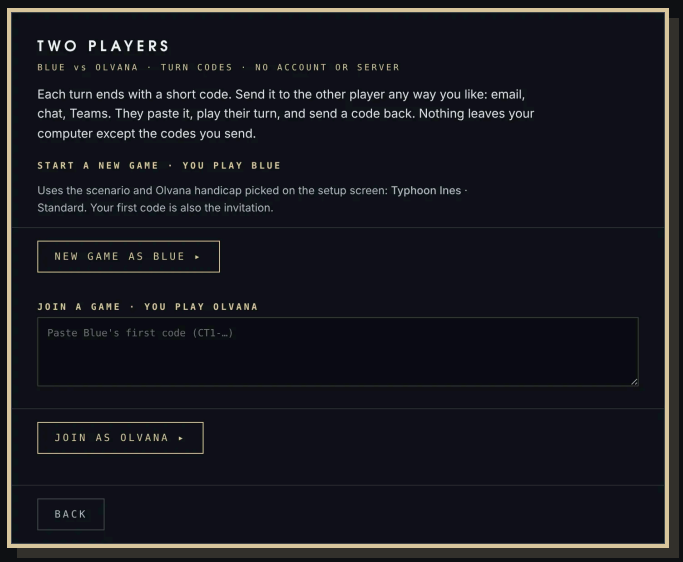


FIG. 17 TWO PLAYERS on the start screen. Blue starts a game; Olvana joins by pasting Blue's first code. Saved games appear under RESUME.

BLUE STARTS

1. Pick a scenario and Olvana handicap on the setup screen.
2. **TWO PLAYERS → NEW GAME AS BLUE.**
3. Play your moves and press **END TURN · GET CODE**.
4. **COPY CODE** and send it. This first code is also the invitation.

OLVANA JOINS

1. **TWO PLAYERS**, paste Blue's code, **JOIN AS OLVANA**.
2. Blue's moves replay on your screen, then it is your turn.
3. End your turn, copy your code, send it back.

THEN, EVERY WEEK

Paste their code, play, send yours. Repeat until week 16. When Olvana's last turn ends the game, Olvana gets one final code to send so Blue's game ends too.

SEND. PASTE. REPEAT.

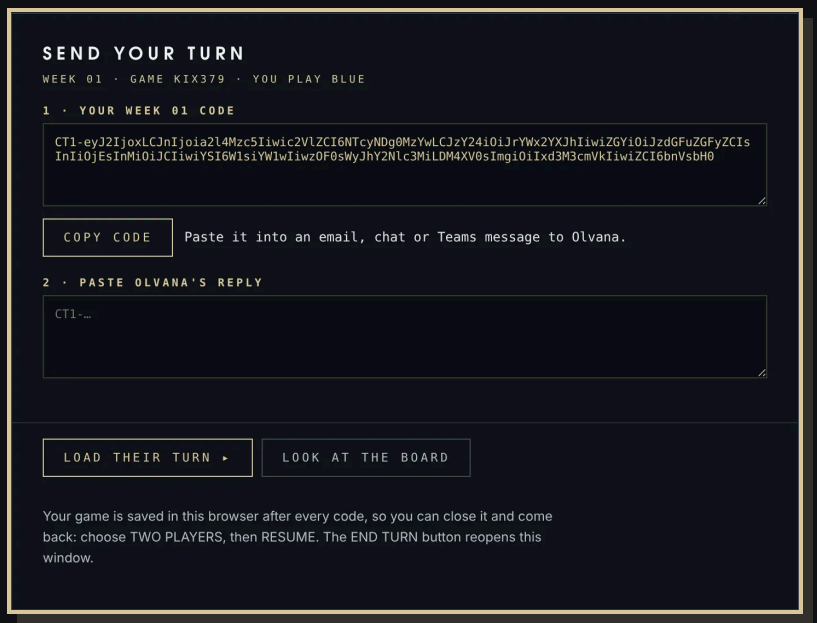


FIG. 18 After END TURN: your code at the top with a COPY button, and a box for your opponent's reply below. LOOK AT THE BOARD closes the window; the END TURN button (now TURN CODES ▸) brings it back.

Q&A

WHAT IS INSIDE A CODE?

Your moves, your new-technology choice, the game's ID and starting seed, and a fingerprint of the board. Nothing personal and nothing from your computer.

DO WE NEED THE INTERNET?

Only to send the code. The game itself runs offline.

CAN WE USE DIFFERENT VERSIONS?

You can mix the Windows app, the Mac app and the HTML file, as long as both of you are on a 0.14 build (0.14, 0.14.1…). For a player-made campaign, both of you import it first.

I CLOSED THE GAME MID-MATCH.

Your game saves after every code. Open **TWO PLAYERS** and pick it under **RESUME**.

"BOARD CHECK FAILED" IN THE LOG?

The two copies disagree. Compare health bars with your opponent; if they differ, restart from the last code you both agree on.

HOUSE RULES

Neither of you sees the other's moves until the code arrives, and new-technology choices stay secret until both are in. That makes it a good fit for a seminar: pair people up, give them a week, and run the debrief together at the end.

WHAT'S DIFFERENT FROM SINGLE PLAYER

- Whoever plays Olvana sees the game from Olvana's side: their brain is "yours", the tabs say PROTECT OLVANA and INFLUENCE BLUE.
- Your own moves get the big result cards; your opponent's appear as small cards at the top of the screen.
- The tutorial is single-player only. Your two-player game stays saved while you take it.

⚙ IN THE TOP BAR · ACCESSIBILITY

MAKE IT YOURS

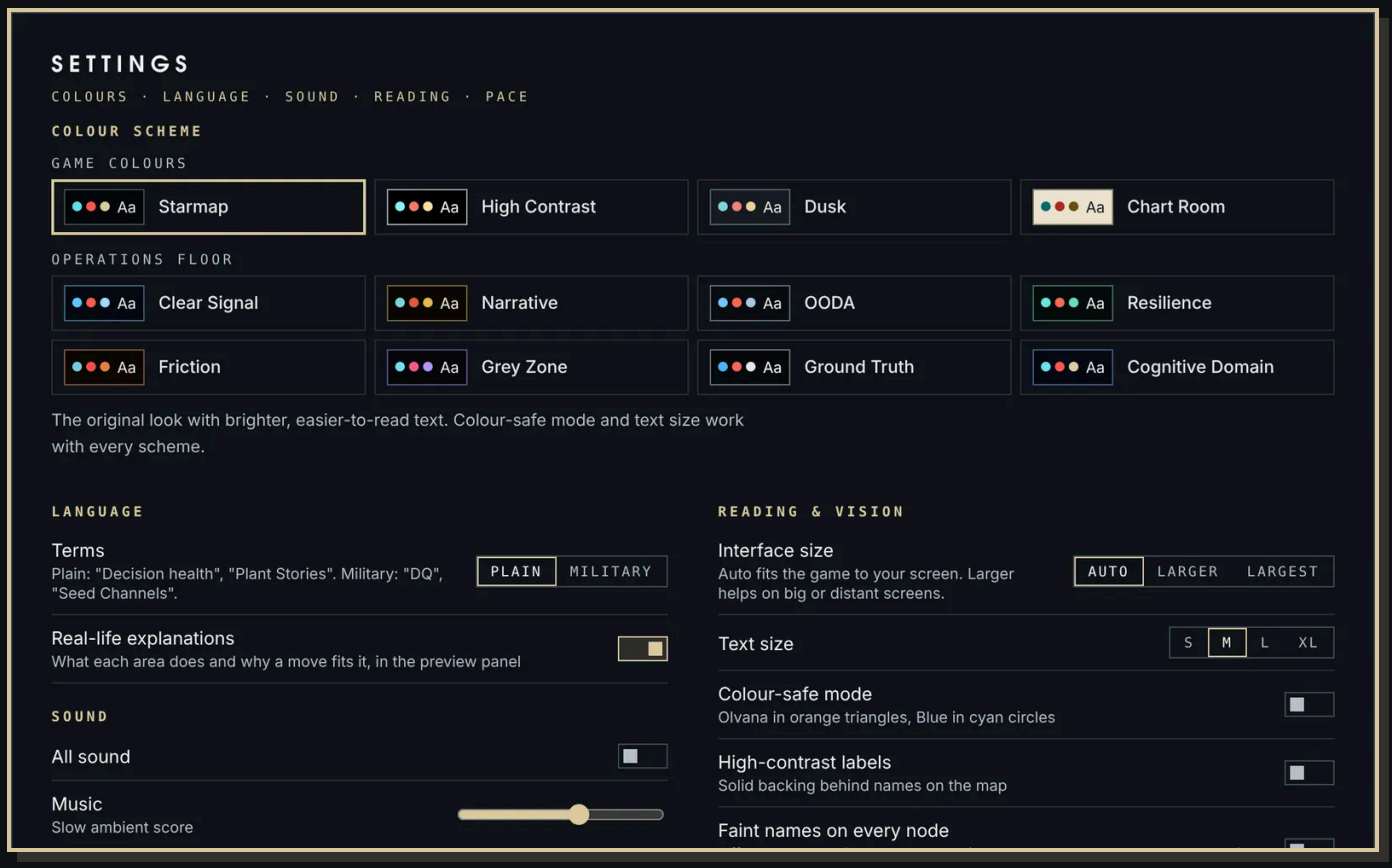


FIG. 19 Settings: twelve colour schemes, language, real-life explanations, sound and music track, interface and text size, colour-safe mode, map labels, the decision-loop overlay, reduced motion, Olvana's turn speed and how often result cards pop up.



FIG. 20 Colour-safe mode: Olvana becomes orange triangles, Blue stays cyan circles, so the sides differ by shape as well as colour.

PLAIN OR MILITARY?

PLAIN	MILITARY
Decision Health	Decision Quality (DQ)
Budget · pts	Influence points · IP
Notice / Interpret	Observe / Orient
Entry point	Gateway
Plant Stories	Seed Channels
Watch Team	Assessment Cell
Crisis	Decision point
What happened	SITREP

In military mode Olvana's moves use its own doctrine names ("Public Opinion Warfare", "Network Swarming").

QUIET PLEASE

Pick from four music tracks in Settings: Night Watch, Undertow, Long Telegram or Dead Air. The music and sound buttons sit in the top bar.

FOUR CAMPAIGNS · ALL ORIGINAL FICTION

CHOOSE YOUR BATTLEGROUND

Same rules, different pressure. Each scenario has its own crises, its own tested areas and its own starting conditions.

TYPHOON INES

DISASTER RELIEF · THE KALVARA ISLANDS

A typhoon flattens a small island republic that trades with both sides. Over sixteen weeks of relief and rebuilding, Kalvara chooses whom to trust.

TWIST None. The best first campaign.

W04	**The Relief Flotilla**	NOTICE
W08	**The Blackout**	INTERPRET
W12	**The False Alarm**	ACT
W16	**The Reconstruction Compact**	TRUST ×2

THE VALMORA VOTE

A SNAP ELECTION · VALMORA

A partner democracy holds a snap election. Olvana quietly favours the candidate who would cancel the Blue trade pact.

TWIST Both TRUST areas start weakened (Blue's more), and AI Synthetic Media arrives first.

W04	**The Leaked Audio**	INTERPRET
W08	**The Ballot Rumor**	NOTICE
W12	**Count Night**	DECIDE
W16	**Transfer of Power**	TRUST ×2

COLLISION AT VARRIN REEF

MARITIME INCIDENT · TELUGA

An Olvanan cutter and a fishing boat from Teluga, a small Blue ally, collide at a disputed reef. Two fishermen are missing.

TWIST Both sides' Government Contacts start weakened, so the fight over envoys starts on day one.

W04	**Two Videos**	NOTICE
W08	**The Missing Crew**	ACT
W12	**The Drills**	ACT
W16	**The Tribunal Ruling**	DECIDE ×2

FEVER IN PORT SABRE

AN OUTBREAK · PORT SABRE

A fast-spreading virus hits a trading city-state that depends on both sides. Rumours move faster than the virus.

TWIST Infodemic. Held nodes push harder on their neighbours and both NOTICE areas start strained.

W04	**Patient Zero**	INTERPRET
W08	**The Bad Batch**	TRUST
W12	**The Forged Order**	ACT
W16	**The Health Accord**	DECIDE ×2



FIG. 21 The campaign picker groups campaigns by opponent. The icons show the area each crisis tests (×2 counts double); the highlighted card is the one you will play.

REPLAY VALUE

Every campaign draws a fresh brain map, and the headlines come from a large story bank written for each scenario, so you can play many games before stories repeat.

TRY THIS

Play the same scenario on **Standard** twice: once attacking every turn, once defending first. Compare the two debriefs.

FOUR CAMPAIGNS VS DONOVIA · ALL ORIGINAL FICTION

A SECOND OPPONENT

NEW IN 0.14

Donovia fights by its own rules (page 25). Its four campaigns are set in fictional countries from the Army's training world; every town, outlet, person and event in them is invented.

WINTER GRID

A COLD SNAP · BOTHNIA

A record cold snap strains the grid of a small northern democracy that buys a third of its winter power from Donovia. Donovia offers to keep the lights on, for a price.

TWIST Blue's FOLLOW-THROUGH starts strained; 4 dormant narratives.

W04	**The First Blackout**	NOTICE
W08	**The Price of Heat**	DECIDE
W12	**The Substation Fire**	ACT
W16	**The Winter Compact**	TRUST ×2

THE LAKE REFERENDUM

A REFERENDUM · FRAMLAND

Framland votes on rebuilding a dam on the lake it shares with Donovia. Donovia wants the joint project; many Framlanders do not.

TWIST Donovia starts with a foothold in Blue's INTERPRET; Blue's TRUST starts weakened.

W04	**The Engineers' Report**	INTERPRET
W08	**Ten Thousand Neighbours**	NOTICE
W12	**Debate Night**	DECIDE
W16	**Referendum Day**	TRUST ×2

THE SILENT SURVEY SHIP

A SHIP GOES QUIET · ARNLAND

An Arnland research ship stops transmitting near Donovian waters, while Arnland decides whether to join Blue's alliance.

TWIST Both DECIDE areas start strained.

W04	**Radio Silence**	NOTICE
W08	**Snap Drills**	ACT
W12	**Three Stories**	INTERPRET
W16	**The Inquiry**	DECIDE ×2

THE VELL VALLEY

ANNEXATION · GORGAS

Donovia has said for years that Donovian speakers in a Gorgas valley need its protection. Now armed men without insignia are on the valley roads.

TWIST 5 dormant narratives; Blue's NOTICE and TRUST start strained.

W04	**The Protection Claim**	INTERPRET
W08	**Men Without Insignia**	ACT
W12	**The People's Vote**	TRUST
W16	**The New Map**	DECIDE ×2

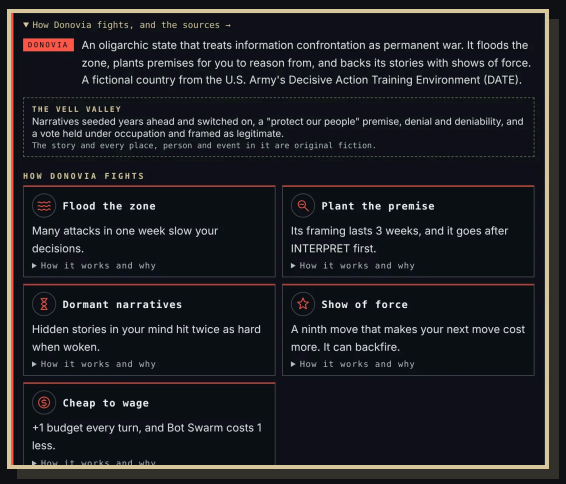


FIG. H How Donovia fights on the start screen: one card per rule.

FICTION, WITH REAL METHODS

No campaign retells a real event. What comes from research is **how Donovia operates**: flooding the zone, planting premises, dormant narratives, shows of force, denial and staged legitimacy. Each campaign names its sources on the start screen (**How Donovia fights**). Donovia, Bothnia, Framland, Arnland and Gorgas are countries in the U.S. Army TRADOC Decisive Action Training Environment (DATE); they are fictional training countries.

CAMPAIGN EDITOR · SHARE · IMPORT

MAKE A CAMPAIGN

NEW IN 0.13.4

Set the scene, place the crises and new technologies on the calendar, tune how strong each side starts, then play it, check its balance and share it.

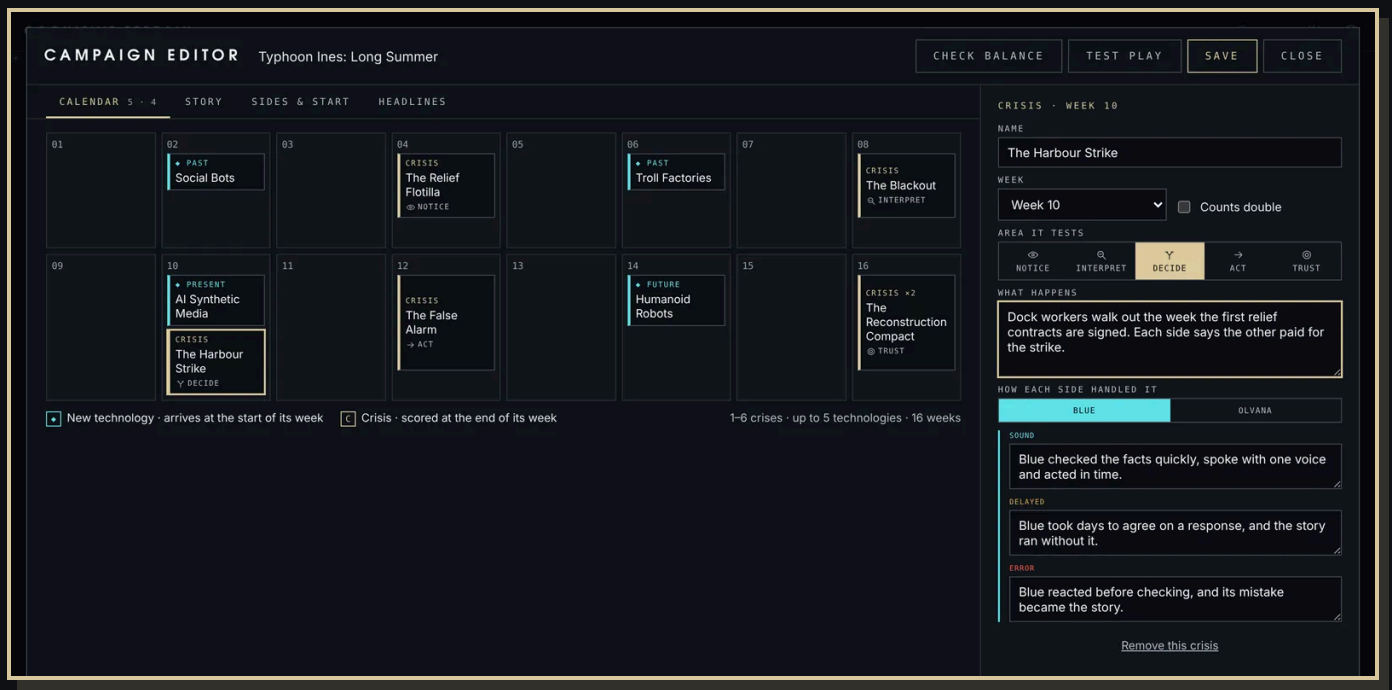


FIG. F The campaign editor, editing a copy of Typhoon Ines with a new crisis in week 10. Click a crisis or technology on the calendar to edit it on the right; + adds one in that week.

MAKE ONE

1. On the start screen press **CAMPAIGNS** (or ⚙ → **CAMPAIGN EDITOR**).
2. **+ NEW CAMPAIGN**, or **COPY & EDIT** a built-in one.
3. Place 1 to 6 crises and up to 5 new technologies. Pick from 18 real past, present and future technologies, or **BUILD YOUR OWN**: its scores, the moves it boosts and by how much. For each crisis write what happens, the area it tests and each side's SOUND, DELAYED and ERROR.
4. **CHECK BALANCE** plays it 40 times computer against computer. **TEST PLAY** starts it.

SHARING

Save a **.campaign.json** file or copy a code that starts `CTC1-`. Campaigns are plain text and cannot run code. For a two-player game, both players import the campaign first; the turn code checks you have the same version.

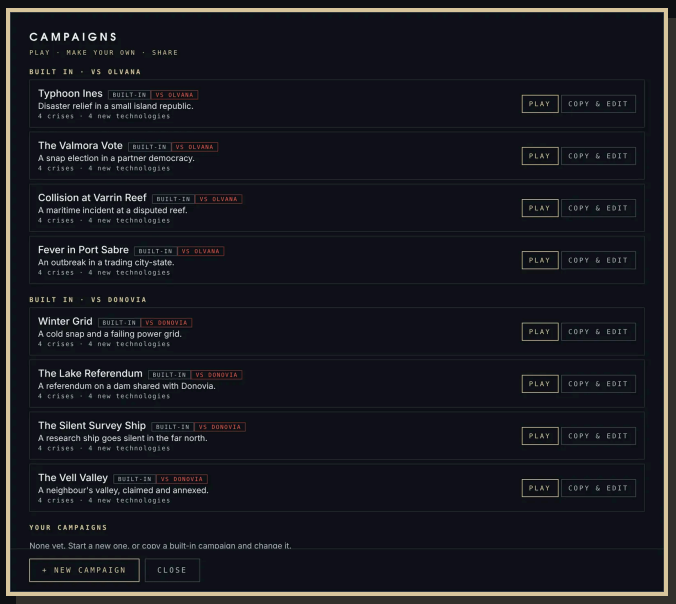


FIG. G The campaign library: play, copy, save as a file, copy a share code, import.

ASYMMETRIC WARFARE

KNOW YOUR ENEMY

Blue is modelled as an open, rule-bound democracy; both opponents as closed, centralized states. They share the rules below, and each adds its own way of fighting: Olvana on page 24, Donovia on page 25.

BOTH OPPONENTS

ALREADY INSIDE	Blue's Social Media entry point starts weakened (80, not 100).
STATE-BACKED CONTENT	Bot Swarm costs 1 less.
RUN FROM THE TOP	Health leans 40% on DECIDE (Blue's: 30%). Fast when intact, brittle when shaken.
SLOW TO LEARN	Lessons Learned heals +2 instead of +4 and builds recovery half as fast.
COSTLY SURGE	An extra move costs 5 points (Blue: 2).

BLUE

OPEN INTERNET	Olvana's posts into Blue spread further.
FREE PRESS & COURTS	Blue's TRUST area recovers faster, and Official Statements hit harder while it holds.
THE TRUTH STICKS	Enemy nodes won over to Blue drift back at about a third of normal speed.
CHEAP SURGE	Blue's extra move costs 2 points; Olvana's costs 5.

TIPS & TACTICS FROM THE WAR ROOM

1. **Open on an entry point.** Plant Stories on their News Media or Social Media gives you the foothold every later move needs.
2. **Hit DECIDE.** Both opponents' centralized command makes it the biggest lever on their health.
3. **Read the top bar.** The next crisis names its area. Defend that area in the two weeks before.
4. **Frame, then plant.** Framed nodes cannot recover, so the second hit captures.
5. **Buy Lessons Learned early.** Faster recovery compounds every week.
6. **Never sit on 18 points.** Surge instead.
7. **Prebunk before, not after.** Half damage is worth more on a cluster that is about to be hit.
8. **Guard TRUST.** It is slow to fall and painful to lose.

HOW OLVANA FIGHTS

Olvana, a fictional country from the Army's training environment, is modelled as a closed, centralized one-party state that wages a long campaign through every arm of government. Each rule says where it comes from; design assumptions are marked as such.

HOW IT FIGHTS

CENSORED INTERNET

Your posts lose reach; censors delete what gets in.

Strongest firewall in the game: your posts into Olvana spread less, 2% more fail outright, and its three entry points recover 20% faster.

DESIGN ASSUMPTION · ATP 3-13.1

STATE-FUNDED CONTENT

Content is cheap and plentiful.

Bot Swarm costs 1 less.

SHIN (NIU)

ALREADY INSIDE

Its personas are in place before the first crisis.

Your Social Media entry point starts at 80, not 100.

SHIN (NIU)

PATIENT AND METHODICAL

Steady pressure, not bursts.

The computer works one of your areas at a time, varies its cards and rarely buys an extra move.

SHIN (NIU) · DESIGNER'S READING

WHERE IT IS WEAK

RUN FROM THE TOP

DECIDE is its single point of failure.

Health leans 40% on DECIDE (Blue's: 30%). Fast when intact, brittle when shaken.

SHIN (NIU)

HARD TO ADMIT MISTAKES

Bad news travels up slowly.

Lessons Learned heals +2 instead of +4 and builds recovery half as fast.

DESIGN ASSUMPTION

COSTLY EXTRA MOVES

It cannot improvise cheaply.

An extra move costs Olvana 5 points; Blue pays 2.

DESIGN ASSUMPTION

Shared with Donovia (page 25): already inside your Social Media entry point, cheaper Bot Swarm, run from the top, slow to learn, extra moves cost 5.

BEATING OLVANA

Open on an entry point early, before censors heal it. **Guard Social Media**: it starts weak. **Then go for DECIDE** and hold it: once the top is shaken, Olvana is slow to recover and cannot surge its way out.

DESIGNER'S NOTE

Olvana should feel like a fortress with one gate. Against a human Olvana, Blue feels a slow squeeze: no fireworks, just patient pressure on one area at a time from someone who was already inside. Playing Olvana, you feel safe behind the firewall but know everything hangs on DECIDE, and that once it cracks you learn too slowly to fix it. It is patience against nerve.

— Kevin Williamson

HOW DONOVIA FIGHTS

Donovia, a fictional country from the Army's training world, is modelled as an oligarchic state at permanent information war. Every rule follows published research.

HOW IT FIGHTS

FLOOD THE ZONE

Many attacks in one week slow your decisions.

More than 2 attacks landing in a week: each extra one takes 0.02 off your health until Donovia's next turn (up to 0.06). Each Lessons Learned level absorbs one more.

NATO C2COE · ISW · WILDE & SHERMAN

PLANT THE PREMISE

Its framing lasts longer, and it aims at how you read events.

Frame the Story keeps a node framed 3 weeks, not 2. The computer goes for INTERPRET first, then DECIDE.

NATO C2COE · ISW · IISS

DORMANT NARRATIVES

Stories seeded long ago, hidden in your mind.

3 hidden in your NOTICE, INTERPRET and TRUST nodes (more in some campaigns). The first attack on one hits twice as hard; a Watch Team removes it.

ISW

SHOW OF FORCE

A ninth move: an act, not a post.

3 points: any FOLLOW-THROUGH or DECIDE node, no foothold, no audience roll. Your next move costs 1 more. Failure backfires on the node and its neighbours.

ISW · IISS

CHEAP TO WAGE

Its information war costs little.

+1 budget every turn; Bot Swarm costs 1 less. The computer repeats what works and buys extra moves often.

ISW

WHERE IT IS WEAK

PREDICTABLE, BRITTLE

Its messages are predictable and its threats can backfire.

Prebunk works better against it (×0.45), its failed attacks push back harder, and your attacks on its TRUST area hit 10% harder.

ISW

LEAKY FIREWALL

Its internet is harder to seal than Olvana's.

Your posts lose a third less reach than into Olvana; its entry points recover only 10% faster.

WILDE & SHERMAN

ONE COMMAND

DECIDE is its single point of failure.

Health leans 40% on DECIDE, as Olvana's does.

NATO C2COE

Shared with Olvana (page 24): already inside your Social Media entry point, slow to learn (Lessons Learned +2), extra moves cost 5.

BEATING DONOVIA

Watch Team early on NOTICE, INTERPRET and TRUST to flush out sleepers. **Buy Lessons Learned** to blunt the flood. **Prebunk** what the next crisis tests. ISW: debunking single stories "only grapples with the tactical level"; defend the premises.

DESIGNER'S NOTE

Donovia should feel loud. Against a human Donovia, Blue feels rushed and off balance: too many stories at once, old narratives going off like mines, a show of force just as you planned your turn. The test is whether you keep your head and defend the premises instead of chasing every claim. Playing Donovia, you feel the thrill of cheap, fast aggression, and the risk when your threats blow back. — Kevin Williamson

JARGON BUSTER

SAY IT IN PLAIN WORDS

TERM	PLAIN MEANING	IN THE GAME
INFORMATION OPERATIONS	Using information to affect what an opponent knows, believes and decides, while protecting your own side.	The whole game
COGNITIVE WARFARE	A long contest over how people perceive, interpret, decide and act.	The whole game
OODA LOOP	Observe, Orient, Decide, Act: the four steps of any decision.	NOTICE, INTERPRET, DECIDE, ACT
MISO / PSYOP	Military Information Support Operations: truthful messages aimed at foreign audiences.	Plant Stories, Frame the Story
PUBLIC AFFAIRS	Official communication with the public and press.	Official Statement
OPSEC	Protecting information an opponent could exploit.	Watch Team
PREBUNKING	Teaching people a manipulation trick before they meet it.	Prebunk
AMPLIFICATION	Making a message louder: bots, troll networks, coordinated sharing.	Bot Swarm
SYNTHETIC MEDIA	AI-generated text, images, audio or video.	New technology events
FRAMING	The angle a story is told from.	Frame the Story
ATTRIBUTION	Deciding, and saying publicly, who was responsible.	INTERPRET area
ENGAGEMENT EFFECTIVENESS	A research score for user interactions per post: shares count most, then comments, likes, views.	Spread grade F to A+
COGNITIVE SUPERIORITY	Staying ahead over time: deciding well, adapting faster, getting results cheaper.	LEAD; the knockout win
ACUTE vs CHRONIC	Fast effects (minutes to days) versus slow ones (weeks to years).	Most areas vs TRUST
GATEWAY	A channel into a society's information space.	Entry points ◇
SITREP	Situation report.	◉ in the log
REFLEXIVE CONTROL	Shaping what an opponent sees so it reaches, by itself, the decision you want.	Donovia's Frame the Story
ODIN / DATE / OPFOR	US Army training environment and the fictional countries and opposing forces it defines.	Olvana, Donovia and their neighbours

SOURCES · ATTRIBUTION · LICENSE

WHERE THIS COMES FROM

The game is built on published research. The rules simplify it; the debrief and the in-game guide point back to it.

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- Roozenbeek, J., & van der Linden, S. (2020). *Breaking Harmony Square: A game that inoculates against political misinformation* (cited in Rushing et al.).

ATTRIBUTION AND LICENSE

This game and manual adapt concepts, the capability taxonomy, the engagement-effectiveness metric (formula, interaction weights and grade scale) and two sets of innovation attribute scores from the three Rushing et al. papers above. Each paper is licensed under Creative Commons Attribution 4.0 International (CC BY 4.0): creativecommons.org/licenses/by/4.0. Papers: arxiv.org/abs/2603.05222 · arxiv.org/abs/2510.15805 · arxiv.org/abs/2510.15801.

Changes made: the material was adapted into game mechanics and rewritten for play. Engagement grades are mapped to effect multipliers invented for this game; the OODA effects, capability families and cognitive superiority definition are expressed as game rules; three of the five innovation score sets are the designer's own estimates; all scenario, narrative and tutorial text is original.

No endorsement: the authors and their institutions have not reviewed or endorsed this game.

HOW THIS GAME WAS MADE

Design and direction: Kevin Williamson. Cognitive Terrain is my design: I conceived the game, built its rules from the research, chose every source, defined both opponents and the campaign briefs, set the balance targets and accessibility standards, and reviewed every rule, campaign and page.

Tools: I used AI-assisted development tools to help implement the code, draft text and run balance tests, working from my specifications; I reviewed, revised or rejected everything they produced. The published sources, not the tools, are the authority for any rule that models real-world behaviour. The art, icons and music are drawn and synthesized by the game's own code; no AI image, voice or music generators were used, and no AI runs while you play.

Olvana and Donovia, and the countries Bothnia, Framland, Arnland and Gorgas, are fictional training countries from the US Army's ODIN DATE environment; they are used here without endorsement. All eight scenarios, their towns, outlets, people and every event in them are original fiction. Any resemblance to real places or events is coincidental. Donovia's rules follow the NATO, ISW, IISS and Carnegie research above; game effects are abstractions of broad, publicly reported patterns, not claims about any specific operation.

CUT OUT AND KEEP

QUICK REFERENCE CARD

EACH TURN

- 3 moves (2 if DECIDE is lost)
- +6 budget, +1 per enemy node held
- Budget caps at 18
- Surge: +1 move for 2 pts (Blue)
- Keys 1–8 pick cards; Esc cancels

CONTROL

- Hold a node at 30+
- Foothold at 10+: reach its links
- Entry points ◇ always in reach
- Half an area held = area lost
- Nodes recover a little each week

CALENDAR

- New technology: weeks 2, 6, 10, 14
- Crises: weeks 4, 8, 12, 16 (16 counts ×2)
- SOUND 3 · DELAYED 1 · ERROR 0
- Knockout: 2 areas ahead, 2 weeks running

CARD	KEY	COST	DOES	BEST IN
PLANT STORIES	1	2	Push 20; ×1.5 on entry points	NOTICE
FRAME THE STORY	2	2	Push 16; no recovery for 2 turns	INTERPRET, DECIDE, TRUST
BOT SWARM	3	3	Push 12 + 7 on every link	ACT, TRUST
REPOST & REBRAND	4	2	Push 10; strips Watched and Prebunked	Anywhere
WATCH TEAM	5	2	3 weeks: attacks weaker, statements ×1.5	NOTICE
PREBUNK	6	2	3 weeks: half influence on node + links	INTERPRET
OFFICIAL STATEMENT	7	2	Restore 14+; half if TRUST is lost	DECIDE
LESSONS LEARNED	8	3	All your nodes +4; faster recovery (max 3)	ACT

LOSE THIS AREA…	…AND THIS HAPPENS
NOTICE	You can no longer see the other brain clearly
INTERPRET	Every move costs +1
DECIDE	−1 move per turn
ACT	1 in 4 moves fails
TRUST	Statements half, recovery half, enemy posts +1 grade


Entry point


Prebunked


Contested


Your node under pressure